Dynamic Late Fusion Multiple Kernel k-means Clustering via Adaptive Sample Selection and Alignment

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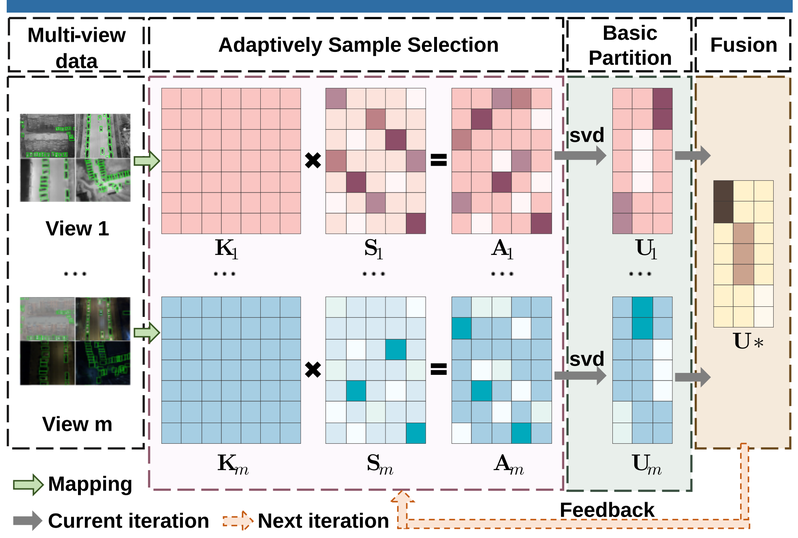
Motivation

Multiple kernel k-means clustering (MKKM) maps samples to high-dimensional kernel spaces and is widely used in multimedia analysis. However, two **structural bottlenecks** persist:

- **Computational wall.** Reconstructing the clustering partition relies on eigendecomposition of an $n \times n$ kernel matrix, incurring $\mathcal{O}(n^3)$ cost that blocks large-scale application.
- **Sampling blind spot.** Anchor/sampling-based MKKM reduces cost but lacks theoretical guarantees on information loss, and is sensitive to initialization. No analysis quantifies how well the sampled partition approximates the full-sample ground truth.

Goal: Investigate a dynamic sampling-based MKKM algorithm that reduces computational complexity and extends MKKM to large-scale data, while providing both theoretical and empirical guarantees for the sampling mechanism.

LF-ASSA Framework



Three modules: adaptive sampling matrices $\{\mathbf{S}_p\} \to$ SVD-based partition reconstruction $\{\mathbf{U}_p\} \to$ aligned late fusion into consensus $\mathbf{U}^*$ with feedback.

Objective (LF-ASSA):

$$\max_{\mathcal{L}} \sum_{p=1}^{m} \Big[\operatorname{Tr}\big(\mathbf{U}_p^\top \mathbf{K}_p \mathbf{S}_p \mathbf{V}_p\big) + \lambda \beta_p \operatorname{Tr}\big(\mathbf{T}_p^\top \mathbf{U}_p^\top \mathbf{U}^*\big) \Big]$$

$$\text{s.t. } \mathbf{U}_p^\top \mathbf{U}_p = \mathbf{I}_k,\ \mathbf{V}_p^\top \mathbf{V}_p = \mathbf{I}_k,\ \mathbf{S}_p^\top \mathbf{S}_p = \mathbf{I}_s,$$
$$\mathbf{T}_p \mathbf{1} = \mathbf{1},\ \mathbf{T}_p^\top \mathbf{1} = \mathbf{1},\ (\mathbf{U}^*)^\top \mathbf{U}^* = \mathbf{I}_k,\ \boldsymbol{\beta} \in \nabla$$

$\mathcal{L} = \{\mathbf{U}_p, \mathbf{V}_p, \mathbf{S}_p, \mathbf{T}_p; \boldsymbol{\beta}, \mathbf{U}^*\}$, $\nabla = \{\boldsymbol{\beta} \in \mathbb{R}^m \mid \sum_p \beta_p^2 = 1, \beta_p \geq 0\}$. First term: novel adaptive-sampling KKM paradigm; second term: doubly-stochastic late-fusion alignment.

Our Contributions

- We propose a novel MKKM named **LF-ASSA**, which alternatively optimizes sampling matrices and clustering partitions to achieve mutual promotion.
- We provide a **theoretical analysis** demonstrating the advantages of our adaptive sampling mechanism over random sampling in terms of **energy capture**.
- Comprehensive experiments across datasets of different scales verify the effectiveness and efficiency of the proposed method; ablation and extended experiments confirm the superiority of the **dynamic sampling mechanism** over conventional approaches.

Alternating Optimization

A six-stage update;

1. $\mathbf{S}_p = \mathbf{P}_s \mathbf{Q}_s^\top$
   top-s left/right singular vectors of $\mathbf{K}_p^\top \mathbf{U}_p \mathbf{V}_p^\top$
2. $\mathbf{U}_p \leftarrow \operatorname{SVD}(\mathbf{D}),\ \mathbf{D} = \mathbf{K}_p \mathbf{S}_p \mathbf{V}_p + \lambda \beta_p \mathbf{U}^* \mathbf{T}_p^\top$
3. $\mathbf{V}_p \leftarrow \operatorname{SVD}(\mathbf{S}_p^\top \mathbf{K}_p^\top \mathbf{U}_p)$
4. $\mathbf{T}_p \leftarrow$ closed form
   $\mathbf{T} = \mathbf{R} + \big(\frac{1}{k}\mathbf{I} + \frac{\mathbf{1}^\top \mathbf{R} \mathbf{1}}{k^2} - \frac{1}{k}\mathbf{R}\big)\mathbf{1}\mathbf{1}^\top - \frac{1}{k}\mathbf{1}\mathbf{1}^\top \mathbf{R},\ \mathbf{R}_p = \mathbf{U}_p^\top \mathbf{U}^*$
5. $\beta_p = \delta_p \big/ \sqrt{\sum_{q=1}^m \delta_q^2},\ \delta_p = \operatorname{Tr}\big((\mathbf{U}^*)^\top \mathbf{U}_p \mathbf{T}_p\big)$
6. $\mathbf{U}^* \leftarrow \operatorname{SVD}\big(\sum_{p=1}^m \beta_p \mathbf{U}_p \mathbf{T}_p\big)$

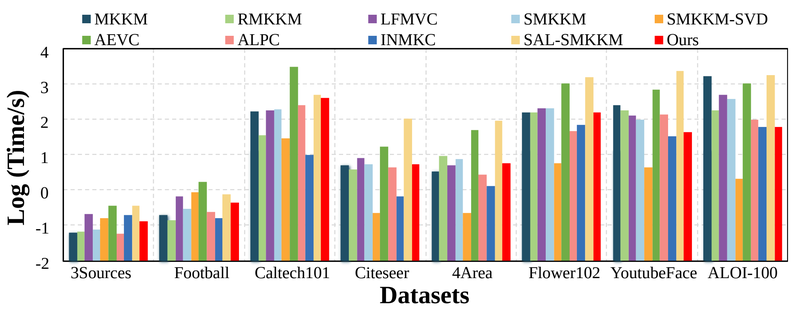
Output: run k-means on $\mathbf{U}^*$ for the final labels.

Clustering Performance (ACC %)

Dataset	LF-ASSA (Ours)	Best Baseline
3Sources	**50.86**	47.34 (LFMVC)
Football	**82.94**	80.79 (LFMVC)
Caltech101	**31.83**	31.38 (LFMVC)
Citeseer	**54.08**	48.35 (LFMVC)
4Area	**75.31**	82.55 (SAL-SMKKM)
Flower102	**42.59**	38.95 (LFMVC)
YoutubeFace	**26.04**	25.15 (LFMVC)
ALOI-100	**67.54**	67.14 (SMKKM)

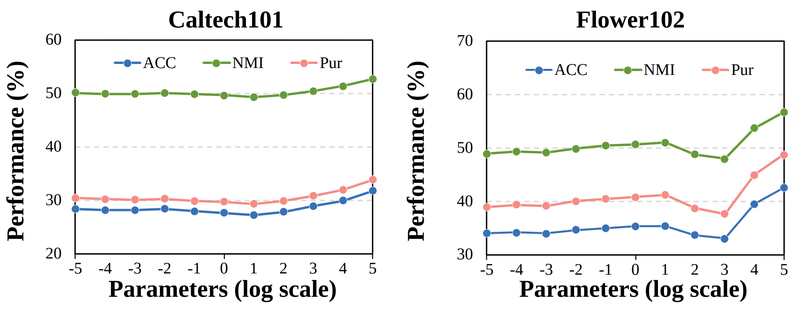
Red: optimal. Blue: sub-optimal. Compared against 9 baselines (MKKM, RMKKM, LFMVC, SMKKM, SAL-SMKKM, SMKKM-SVD, AEVC, ALPC, INMKC).

Efficiency



- Average runtime on small datasets; **top-3 on large-scale** ($n \geq 10{,}000$).
- **Fastest on ALOI-100** among all compared methods.
- Two-step iterative SVD is cheaper than a single full eigendecomposition $\Rightarrow$ scalable to large n.

Hyperparameter Sensitivity



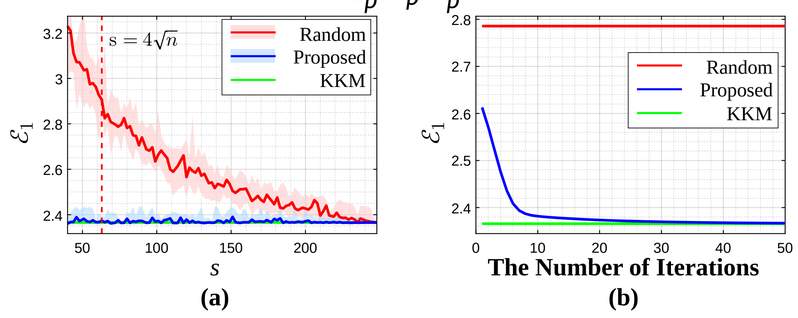
Robust to the balance parameter λ over a wide range (e.g. Caltech101); a performance trough appears on Flower102 at $\lambda = 2^3$.

Adaptive Sampling (Single-View)

We learn a view-specific sampling matrix $\mathbf{S}_p \in \mathbb{R}^{n \times s}$ so the selected anchors $\mathbf{A}_p = \mathbf{K}_p \mathbf{S}_p$ adapt to the data distribution:

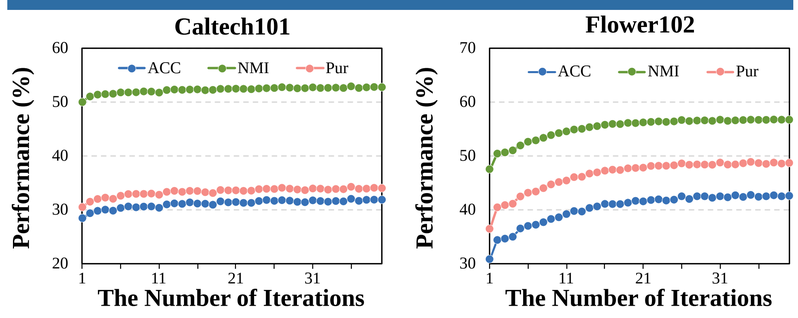
$$\max_{\mathbf{U}_p, \mathbf{V}_p, \mathbf{S}_p} \operatorname{Tr}\big(\mathbf{U}_p^\top \mathbf{K}_p \mathbf{S}_p \mathbf{V}_p\big)$$
$$\text{s.t. } \mathbf{U}_p^\top \mathbf{U}_p = \mathbf{I}_k,\ \mathbf{V}_p^\top \mathbf{V}_p = \mathbf{I}_k,\ \mathbf{S}_p^\top \mathbf{S}_p = \mathbf{I}_s.$$

Closed form via truncated SVD: $\mathbf{S}_p = \mathbf{P}_s \mathbf{Q}_s^\top$, where $\mathbf{P}_s, \mathbf{Q}_s$ are the top-s left / right singular vectors of $\mathbf{K}_p^\top \mathbf{U}_p \mathbf{V}_p^\top$.



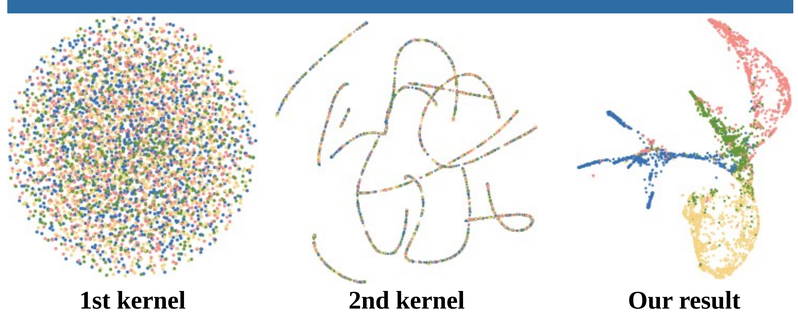
Adaptive sampling (blue) stays near the full-sample KKM baseline as s grows, while random sampling (red) decays with wider variance $\Rightarrow$ robust to initialization.

Convergence



Clustering performance evolves and stabilizes within 40 iterations; minor overfitting is suppressed by the stopping criterion.

Cluster Visualization (t-SNE)



*t-SNE of two kernel views vs. LF-ASSA partition on **4Area**: the learned $\mathbf{U}^*$ reveals clean cluster structure.*

Ablation Study (ACC %)

Method	L1	L2	L3	L4	L5	Ours
Football	61.27	71.39	**82.04**	80.65	75.14	**82.94**
Citeseer	40.67	38.46	39.48	**42.63**	25.61	**54.08**
Flower102	32.52	32.45	40.59	**40.88**	40.18	**42.59**
ALOI-100	66.00	**66.21**	65.89	66.00	65.04	**67.54**

- **L5** (fixed random sampling) is consistently the worst variant $\Rightarrow$ **dynamic adaptive sampling** is a key contributor; every component (L1–L4) adds gains.

Conclusion

- **LF-ASSA** jointly optimizes adaptive sampling and aligned late fusion.
- Replaces eigendecomposition with two SVDs $\Rightarrow$ scalable, with theoretical energy-capture guarantees over random sampling.
- Ablation confirms every component (esp. dynamic sampling, L5) matters.